

## Project Design Phase-II Data Flow Diagram & User Stories

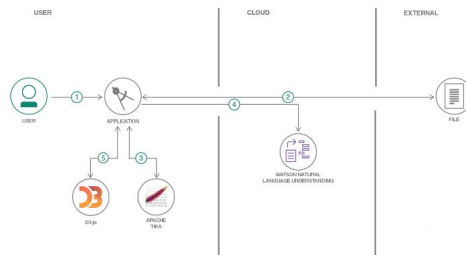
|               |                                                                                                 |
|---------------|-------------------------------------------------------------------------------------------------|
| Date          | 04 July 2026                                                                                    |
| Team ID       |                                                                                                 |
| Project Name  | <b>Comprehensive Analysis Dietary Strategies with Tableau:A college food choices case study</b> |
| Maximum Marks | 4 Marks                                                                                         |

### Data Flow Diagrams:

A Data Flow Diagram (DFD) is a traditional visual representation of the information flows within a system. A neat and clear DFD can depict the right amount of the system requirement graphically. It shows how data enters and leaves the system, what changes the information, and where data is stored.

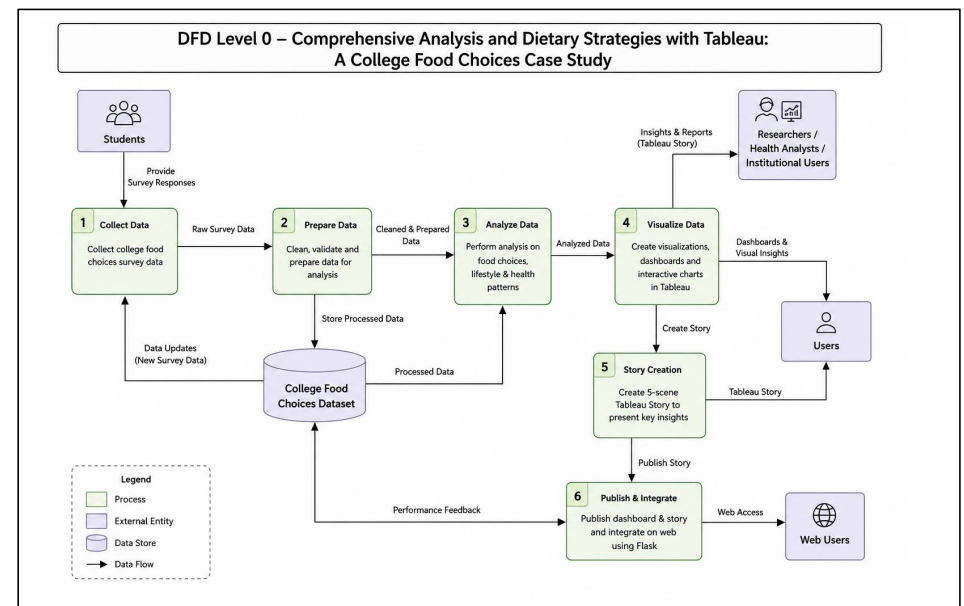
### Example: (Simplified)

Flow



1. User configures credentials for the Watson Natural Language Understanding service and starts the app.
2. User selects data file to process and load.
3. Apache Tika extracts text from the data file.
4. Extracted text is passed to Watson NLU for enrichment.
5. Enriched data is visualized in the UI using the D3.js library.

### Example: DFD Level 0 (Industry Standard)



## User Stories

Use the below template to list all the user stories for the product.

| User Type                  | Functional Requirement (Epic) | User Story Number | User Story / Task                                                                                 | Acceptance criteria                                             | Priority | Release  |
|----------------------------|-------------------------------|-------------------|---------------------------------------------------------------------------------------------------|-----------------------------------------------------------------|----------|----------|
| Project Team               | Data Collection & Preparation | USN-1             | As a team member, I can collect and connect the College Food Choices dataset to Tableau.          | Dataset is successfully loaded and available in Tableau.        | High     | Sprint-1 |
| Project Team               | Data Collection & Preparation | USN-2             | As a team member, I can validate the dataset before analysis.                                     | Records and fields are verified for analysis.                   | High     | Sprint-1 |
| Project Team               | Data Preparation              | USN-3             | As a team member, I can clean and prepare the data for visualization.                             | Required null values and data issues are handled.               | High     | Sprint-2 |
| Project Team               | Data Visualization            | USN-4             | As a user, I can view visualizations of student food choices and lifestyle patterns.              | Charts display the selected data correctly.                     | High     | Sprint-1 |
| Student / User             | Data Visualization            | USN-5             | As a user, I can explore breakfast, comfort food, and healthy-feeling patterns.                   | Relevant visualizations display the required patterns.          | High     | Sprint-2 |
| Student / User             | Dashboard                     | USN-6             | As a user, I can view key dietary and lifestyle insights on an interactive dashboard.             | Dashboard displays multiple visualizations correctly.           | High     | Sprint-1 |
| Student / User             | Dashboard                     | USN-7             | As a user, I can filter dashboard insights by Gender, Breakfast, and Healthy Feeling.             | Selected filters update the visualizations correctly.           | High     | Sprint-2 |
| Student / User             | Story                         | USN-8             | As a user, I can view project insights through a structured Tableau Story.                        | The <b>7-scene Tableau Story</b> displays insights in sequence. | High     | Sprint-2 |
| Researcher/ Health Analyst | Story                         | USN-9             | As a researcher, I can understand major dietary and lifestyle patterns through the project story. | Insights are presented clearly through Story scenes.            | Medium   | Sprint-1 |

| User Type               | Functional Requirement (Epic)         | User Story Number | User Story / Task                                                                                    | Acceptance criteria                                     | Priority | Release  |
|-------------------------|---------------------------------------|-------------------|------------------------------------------------------------------------------------------------------|---------------------------------------------------------|----------|----------|
| Project Team            | Performance Testing                   | USN-10            | As a team member, I can verify that all data records are rendered correctly.                         | Data is successfully loaded without major errors.       | High     | Sprint-1 |
| Project Team            | Performance Testing                   | USN-11            | As a team member, I can verify filters, calculated fields, and visualizations.                       | Filters and visualizations function correctly.          | High     | Sprint-2 |
| Web User                | Web Integration                       | USN-12            | As a web user, I can access the published Tableau dashboard.                                         | Dashboard is available through the web interface.       | High     | Sprint-1 |
| Web User                | Web Integration                       | USN-13            | As a web user, I can access the embedded dashboard and Story through the project interface.          | Dashboard and Story load successfully in the interface. | Medium   | Sprint-2 |
| Project Team / Reviewer | Project Demonstration & Documentation | USN-14            | As a reviewer, I can understand the complete project through documentation and an explanation video. | Documentation and project demonstration are completed.  | High     | Sprint-1 |